Numerical and Temporal Fact Checking

1 Introduction

With the growth of digital tools for disseminating information, misinformation, and disinformation has also increased tremendously in recent years. The misinformation has serious consequences, such as initiating turmoil [11] in society. While numerous fact-checking efforts have been undertaken by websites like Politifact, snopes, Africa check, manual checking of growing misinformation is a cumbersome task.

Example: Claim from QuanTemp

Claim: `Under GOP plan, U.S. families making $86k see avg tax increase of $794.`

[Evidence]: If enacted, the Republican tax reform proposal would saddle only 8 million households that earn up to $86,100 with an average tax increase of $794 …. Only a small percentage (6.5 percent) of the nearly 122 million households in the bottom three quintiles will actually face a tax increase.

[Verdict]: False

Figure 1: Example claim from QuanTemp

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The case of real-world numerical claims Numerical claims are more important to verify as by the Illusion-of-Numeric-Truth effect [3], people tend to believe claims grounded in numbers more even if they are false. Individuals often manipulate numbers and add abstract logic to create a mirage of truth to make narratives in their favor. However, such claims can have various negative impacts on society. For instance, the opioid pandemic in America was believed to be caused by unsubstantiated, exaggerated claims made regarding the effectiveness of the drug [9]. In this project, we define numerical claims as those which *contain explicit or implicit numerical/temporal information and may require numerical operations for verification.*

Challenges in verifying numerical claims Verification of numerical claims requires numerical reasoning ability which involves understanding numeric patterns and mathematical concepts like arithmetic, numerical estimation, and data interpretation. While several works focus on automated verification of natural claims, only a handful of works[2, 5, 10, 6, 4, 1] focus on claims that require numerical reasoning. However, these techniques only focus on the detection of numerical claims[4][5] or are restricted to specific statistical properties[6][10]. Existing approaches handle numerical elements in natural claims as regular tokens without attempting to understand their significance.

A typical fact verification pipeline consists of three stages: claim detection, evidence retrieval, and reranking and veracity prediction.

The veracity predictions is usually performed using a fine-tuned model for Natural language Inference.

Numerical claims verification poses a unique challenge, where a fact-checking system must critically analyze and reason about the numerical data presented in both the claim and its evidence. For example, in verifying the claim shown in Figure 1 as 'False', the NLI model needs to identify that the evidence only mentions 8 million households with incomes up to $86k facing tax increases, contradicting the claim of tax increases for all families earning $86k.

In this project, the goal would be to improve the NLI model for veracity prediction in fact-checking pipeline with focus on numerical claims.

2 Resources

You will be working on the QuanTemp dataset collected for the purpose of numerical fact-checking. The dataset comprises 15K claims with train, val and test splits and an evidence collection with 420K snippets. The paper related to the dataset can be found at [7]. The data for this can be found at https://github.com/factiverse/QuanTemp. The repository also has a script that demonstrates off the shelf retrieval and veracity prediction. You can use the Table in NumTemp [7] as template for metrics and format in which results need to be reported. Additionally also report performance on the evaluation set at https://github.com/factiverse/factIR provided with evidence collection. You could use the models trained on Quantemp and evaluate on FactIR [8] evaluation set. Explore different NLI models to improve reasoning and also claim decomposition approaches to improve ranking and downstream verification.

In your report detail the intuition behind the performance changes and also qualitative analysis of few prominent examples. Also detail how the experiments answer the various research questions.

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